

ACoT-VLA: Action Chain-of-Thought for Vision-Language-Action Models

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Abstract

Vision-Language-Action models have emerged as essential generalist robot policies for diverse manipulation tasks, conventionally relying on directly translating multimodal inputs into actions via Vision-Language Model embeddings. Recent advancements have introduced explicit intermediary reasoning—such as sub-task prediction (language) or goal image synthesis (vision)—to guide action generation. However, these intermediate reasoning are often indirect and inherently limited in their capacity to convey the full, granular information required for precise action execution. Instead, we posit that the most effective form of reasoning is one that deliberates directly in the action space. We introduce **Action Chain-of-Thought (ACoT)**, a paradigm where the reasoning process itself is formulated as a structured sequence of coarse action intents that guide the final policy. In this paper, we propose ACoT-VLA, a novel architecture that materializes the ACoT paradigm. Specifically, we introduce two complementary components: an **Explicit Action Reasoner (EAR)** and **Implicit Action Reasoner (IAR)**. The former proposes coarse reference trajectories as explicit action-level reasoning steps, while the latter extracts latent action priors from internal representations of multimodal input, co-forming an ACoT that conditions the downstream action head to enable grounded policy learning. Extensive experiments in real-world and simulation environments demonstrate the superiority of our proposed method. Code is available at: <https://github.com/AgibotTech/ACoT-VLA>.

1. Introduction

To overcome the generalization limits of task-specific robot policies [11, 48, 63], recent work has converged on Vision-Language-Action (VLA) models [5, 24, 33, 41], which always leverage a pre-trained Vision-Language Model (VLM) [1, 2, 46] to encode visual and linguistic inputs into a latent representation that conditions an action decoder.

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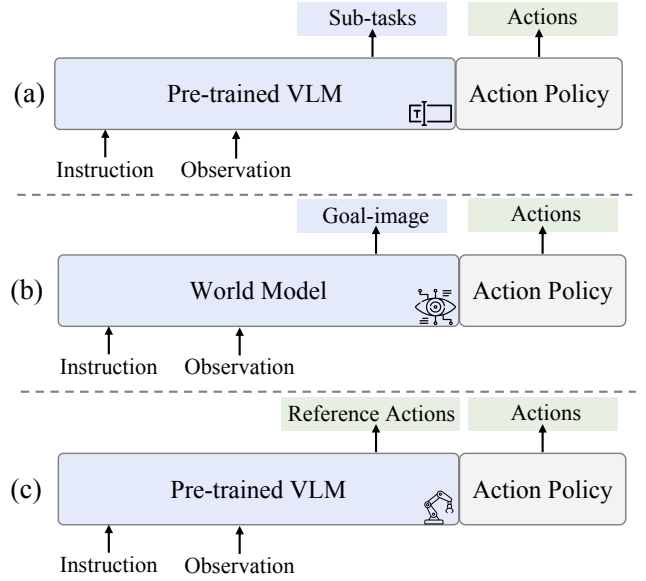


Figure 1. Chain-of-Thought in different space. (a) Language CoT paradigm predicts sub-tasks as intermediate reasoning. (b) Visual CoT paradigm synthesizes a goal image to provide guidance for action policy. (c) Our proposed Action CoT directly operates in action space and provides homogeneous action guidance.

Recent advancements seek to improve the mapping from the input space to the action space by introducing the intermediate reasoning step by language generation, leading to more generalized and precise action outputs [22, 55], as visualized in Fig. 1(a). A parallel thrust leverages world models [16, 51, 67] to simulate environmental dynamics, directly enhancing the efficacy and goal-oriented nature of the generated action sequences [60, 64], as shown in Fig. 1(b).

Despite the promising trajectory set by these paradigms, a critical challenge persists: existing generalist policies think predominantly in the vision-language (input) space, often failing to adequately address the inherent disparity between these rich, semantic representations and the requirements of precise, low-level action execution (output). Specifically, the knowledge encoded within the VLM backbone of VLA models is derived from pre-training on web-scale datasets focused on semantic alignment and question-

answering, yielding representations optimized for linguistic understanding rather than physical dynamics. Similarly, while world models forecast future visual states conditioned on inputs, their guidance remains tethered to naturally visual representations. Crucially, both semantic and visual forms of reasoning only offer suboptimal, indirect guidance for generating the necessary action sequence. Consequently, these prevailing approaches rely on an inherently constrained information conduit, struggling to convey the full, granular knowledge of the action space essential for truly grounded and accurate robotic policy learning.

The inherent semantic-kinematic gap in existing policies, *i.e.*, a fundamental disconnect between high-level, abstract inputs and low-level, executable motor commands, necessitates a paradigm shift in how guidance is provided. We contend that to bridge this chasm, policies require guidance that is kinematically coherent, rather than purely semantic or visual. This core principle underpins our novel framework: **Action Chain-of-Thought (ACoT)** (Fig. 1(c)). We redefine the “thought” process not as a sequence of linguistic tokens, but as a structured chain of explicit, kinematically-grounded action intents. This approach furnishes the policy with direct motion cues, supplanting indirect representations. In a manner analogous to learning from physical demonstration, this direct conditioning on action-space information enables a substantially more efficient and veridically grounded policy learning process.

This foundational shift, however, introduces a critical and distinct research challenge: *“How can we robustly and efficiently synthesize the complex, high-dimensional motion cues required for ACoT reasoning from the raw, heterogeneous multimodal inputs?”*

Action-related information manifests in two complementary forms, *i.e.*, explicit or implicit. The explicit form corresponds to observable motion trajectories, such as those in human demonstrations, which directly encode executable patterns of behavior. In contrast, the implicit form resides in latent cues, *e.g.*, linguistic expressions like “reach out” or “grasp”, as well as interaction intents embedded in visual contexts. Although these cues are not presented as explicit robotic trajectories, they implicitly define distributions over feasible actions within the action space. Building upon this insight, we introduce two synergistic mechanisms to generate both explicit and implicit guidance in the action space. We first propose the Explicit Action Reasoner (EAR), which is realized as a light-weight transformer. Particularly, EAR synthesizes coarse-grained motion trajectories conditioned on multimodal observations, offering direct and executable guidance within the action space. Secondly, we devise the Implicit Action Reasoner (IAR), which infers latent action priors through applying cross-attention modeling between downsampled multimodal representations and learnable queries, thereby providing implicit behavioral pri-

ors. Note that these two mechanisms are inherently complementary to each other. Subsequently, through jointly leveraging both EAR and IAR, we develop ACoT-VLA, an integrated Action Chain-of-Thought framework that enables grounded generalist robot policy learning. Extensive experiments across both real-world settings and three simulation benchmarks consistently demonstrate the effectiveness and versatility of our ACoT-VLA.

To summarize, our main contributions are as follows:

- Conceptually, we introduce Action Chain of Thought (ACoT), a new paradigm for generalist robot policies. To the best of our knowledge, this is the first work to formulate the deliberative process as a structured chain of explicit action-space intents, rather than abstract linguistic or visual sub-goals.
- We delve into essential action space guidance and propose the Explicit and Implicit Action Reasoners, which provide both explicit trajectory guidance and implicit behavioral inspiration for action prediction.
- Building upon these two modules, we further propose ACoT-VLA, a unified framework for grounded generalist robot policy learning.
- Empirically, we validate our approach through extensive simulation and real-world experiments, achieving state-of-the-art performance on multiple benchmarks.

2. Related Works

Vision-Language-Action Models. VLA models [14, 18, 19, 27] incorporate pre-trained VLM models to predict language-driven robotic action sequences. Early works [24, 69] formulate robot control as an autoregressive sequence generation problem, discretizing continuous actions into bins. Inspired by generative modeling [31, 38, 68], increasing works [5, 20, 33] adopt diffusion-based action policies to synthesize smooth and high-quality action trajectories. Given that robotic manipulation inherently occurs in three-dimensional space, a line of studies [30, 54, 61] have sought to enhance the spatial reasoning capability of VLA models by integrating 3D priors. For instance, SpatialVLA [41] integrates spatial embeddings to endow model with 3D awareness, while 4D-VLA [57] incorporates both spatial and temporal information to enrich representations. Besides, due to the scarcity of large-scale real-world robot demonstrations, a series of efforts [6, 10, 12, 23, 37, 59] focus on data-centric solutions, constructing large-scale robotic datasets through simulation or real-world collection to scale up policy learning. Moreover, recent large-scale co-training approaches, such as $\pi_{0.5}$ [22], GenieReasoner [35] and Gemini Robotics [47], demonstrate the potential of unifying web-scale language understanding with action learning, enhancing the policy’s generalization ability while retaining the reasoning capability of pre-trained foundation models.

World-Model-based Policies. Advances in world mod-

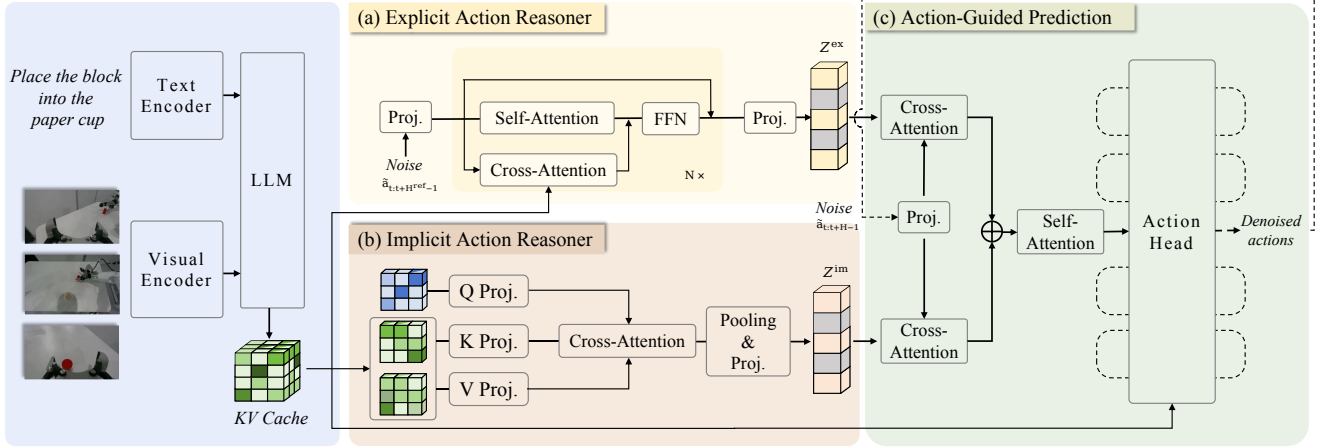


Figure 2. Architectural Overview of ACoT-VLA. The framework consists of three main components operating on features from a shared VLM backbone. (a) The Explicit Action Reasoner (EAR) is a Transformer-based module that synthesizes a coarse reference trajectory, providing explicit action-space guidance. (b) The Implicit Action Reasoner (IAR) employs a cross-attention mechanism with learnable queries to extract latent action priors from the VLM’s internal representations. (c) The Action-Guided Prediction (AGP) head synergistically integrates both explicit and implicit guidances via cross-attention to condition the final denoising process, producing the executable action sequence.

els have demonstrated remarkable capability in synthesizing high-fidelity images and temporally coherent videos. Building upon such progress, emerging researches [26, 29, 36, 58] exploit their predictive dynamics to implicitly guide action generation. Specifically, CoT-VLA [62] introduces visual chain-of-thought reasoning by forecasting sub-goal images, explicitly integrating visual reasoning into action prediction. WorldVLA [8] employs an autoregressive architecture that unifies perception and action generation within a single framework. DreamVLA [60] extends beyond visual prediction and enriches world modeling with dynamic, depth, and semantic cues, improving the model’s physical consistency. Collectively, existing world-model-based methods adopt knowledge-forecasting perspective, incorporating primarily visual guidance into trajectory generation.

In contrast to previous works focusing on visual or linguistic intermediaries for robotic policy learning, our key insight lies in investigating guidance directly within the action space, which intrinsically mitigates the heterogeneity between perception and action, enabling the model to effectively learn action-relevant priors.

3. Methodology

In this section, we present a detailed investigation into how to generate effective action space guidance and integrate it into robotic policy learning. We first define the robotic manipulation problem and formulate our proposed approach in Sec. 3.1. The core of our method lies in two distinct action reasoners introduced in Sec. 3.2 and Sec. 3.3, which provide explicit and implicit guidance within the action space. We conclude by illustrating the policy prediction strategy that effectively integrates this action guidance during pol-

icy learning (Sec. 3.4).

3.1. Problem Formulation

Given a natural language instruction l and current visual observation o_t , the generalist robot policy π_θ aims to predict action sequences $a_{t:t+H-1}$ that accomplishes the specified task. The process can be formally expressed as:

$$a_{t:t+H-1} = \pi_\theta(o_t, l), \quad (1)$$

where H represents the action horizon. Numerous works introduce additional guidance signals g , which encapsulates various forms of auxiliary information to enhance policy’s prediction ability. Specifically, these guidance signals can be broadly categorized into two types: language-level guidance g_{lang} and vision-level guidance g_{vis} . The former is primarily adopted by VLM-based methods, *e.g.*, leveraging LLMs’ reasoning capabilities to predict sub-tasks, while the latter is always employed by world-model-based approaches, such as simulating future observations. Such relationship can be formulated as:

$$\pi_\theta(a_{t:t+H-1}, g \mid o_t, l) = \pi_\theta(a_{t:t+H-1} \mid o_t, l, g) \pi_\theta(g \mid o_t, l), \quad (2)$$

where $g \in \{g_{\text{lang}}, g_{\text{vis}}\}$. Conversely, we shift the focus toward the action domain itself and investigate cues operating directly in the action space, symbolized as g_{action} . The above guidances are extended as $g \in \{g_{\text{lang}}, g_{\text{vis}}, g_{\text{action}}\}$.

Such action guidance can intuitively be disentangled into explicit and implicit forms. The explicit guidance $g_{\text{action}}^{\text{ex}}$ provides direct priors in the form of reference action sequences, whereas the implicit guidance $g_{\text{action}}^{\text{im}}$ arises from contextual signals, *e.g.*, action distribution inherently implied in linguistics.

3.2. Explicit Action Reasoner

To incorporate explicit action trajectories into the thinking process of π_θ to generate high-quality action predictions, we propose the Explicit Action Reasoner (EAR).

We design a mechanism that enables the model to autonomously synthesize reference action sequences as internal guidance for policy learning. Analogously, this formulation can be viewed as an action-space transfer of self-conditioning in generative models [9, 34], where incorporating prior estimates into the generation process has been shown to markedly improve sample quality. Building upon this principle, we instantiate EAR as a light-weight transformer, as shown in Fig. 2 (a), generating kinematically plausible action reference as explicit action-space guidance $\mathcal{G}_{\text{action}}^{\text{ex}}$ for downstream action policy.

Formally, given visual observation o_t and language instruction l , a pre-trained VLM encodes them into a contextual key-value cache:

$$(K_{1:N}^{\text{VLM}}, V_{1:N}^{\text{VLM}}) = \text{VLM}(o_t, l), \quad (3)$$

where N represents the number of layers of VLM. Subsequently, the EAR, denoted as π_θ^{ref} , takes a noisy action sequence $\tilde{a}_{t:t+H^{\text{ref}}-1}$ as input, where H^{ref} indicates the horizon of reference actions. The sequence is first embedded into an initial hidden representation h_0^{ref} , which serves as the input to EAR’s transformer layers. At each transformer layer i , we adopt self-attention, along with cross-attention with the contextual key-value cache from the corresponding VLM layer:

$$\tilde{h}_i^{\text{ref}} = \text{Self-Attn}(h_{i-1}^{\text{ref}}) + \text{CrossAttn}(h_{i-1}^{\text{ref}}, K_i^{\text{VLM}}, V_i^{\text{VLM}}), \quad (4)$$

where self-attention module captures temporal dependencies within the action sequence and cross-attention mechanism injects multimodal contextual priors from the VLM. Then, the intermediate representation $\tilde{h}_i^{\text{ref}}$ is processed by a feed-forward network (FFN) in a residual-parallel manner, updating the i -th EAR representation h_i^{ref} :

$$h_i^{\text{ref}} = h_{i-1}^{\text{ref}} + \text{FFN}(\tilde{h}_i^{\text{ref}}). \quad (5)$$

Through training via flow matching, π_θ^{ref} learns a distribution over action trajectories, producing a denoised action sequence:

$$a_{t:t+H^{\text{ref}}-1}^{\text{ref}} = \pi_\theta^{\text{ref}}(\tilde{a}_{t:t+H^{\text{ref}}-1}, K_{1:N}^{\text{VLM}}, V_{1:N}^{\text{VLM}}). \quad (6)$$

The generated sequence is then encoded via a MLP projector to obtain action embedding Z^{ex} , which serves as explicit action-space guidance $\mathcal{G}_{\text{action}}^{\text{ex}}$ for action policy learning.

3.3. Implicit Action Reasoner

Beyond the explicit action trajectories, the multimodal latent space of VLM also encodes implicit motion cues [13,

40], e.g., visual affordances and action-related semantics. Effectively extracting these action-relevant representations potentially offers complementary guidance. To this end, we introduce an Implicit Action Reasoner (IAR), which directly operates on the VLM’s key-value cache.

Concretely, as presented in Fig. 2 (b), for each VLM layer $i \in [1, N]$, we initialize a learnable matrix $Q_i \in \mathbb{R}^{M \times d}$, where M is a hyperparameter and d represents VLM’s hidden dimension. Considering the information redundancy within VLM’s key-value cache and computational efficiency, we first downsample the corresponding key-value pairs into a lower-dimensional space, which is formulated as:

$$Q'_i = Q_i W_Q^{(i)}, \quad K'_i = K_i^{\text{VLM}} W_K^{(i)}, \quad V'_i = V_i^{\text{VLM}} W_V^{(i)}, \quad (7)$$

where $W_Q^{(i)}, W_K^{(i)}, W_V^{(i)} \in \mathbb{R}^{d \times d'}$ are learnable linear projectors and $d' \ll d$.

Later, cross-attention is applied to extract action-relevant information from each K'_i and V'_i . The resulting features are subsequently integrated via average pooling and transformed through a MLP projector, as visualized in Fig. 2 (b), producing compact representations that capture the implicit action semantics z_i^{im} embedded in VLM’s i -th layer:

$$z_i^{\text{im}} = \text{MLP}(\text{Pool}(\text{CrossAttn}(Q'_i, K'_i, V'_i))). \quad (8)$$

Then, through aggregating these representations across layers, we obtain implicit action-related feature Z^{im} , which serves as implicit action-space guidance $\mathcal{G}_{\text{action}}^{\text{im}}$, complementing the explicit motion priors.

3.4. Action-Guided Prediction

Building upon the explicit action embedding Z^{ex} produced by EAR and implicit action-related feature Z^{im} obtained in IAR, in this section, we introduce the Action-Guided Prediction (AGP) strategy to incorporate both action guidances into policy learning.

As illustrated in Fig. 2 (c), given a noisy action segment $\tilde{a}_{t:t+H-1}$, we first encode it into noisy action embedding via a MLP projector. Particularly, unlike previous approaches that directly feed this embedding into action head π_θ^{head} , we treat it as action query, denoted as Q_{action} , which interacts with both Z^{ex} and Z^{im} to retrieve complementary priors for conditional prediction.

Specifically, we perform dual cross-attention operations:

$$S^{\text{ex}} = \text{CrossAttn}(Q_{\text{action}}, Z^{\text{ex}}, Z^{\text{ex}}), \quad (9)$$

$$S^{\text{im}} = \text{CrossAttn}(Q_{\text{action}}, Z^{\text{im}}, Z^{\text{im}}), \quad (10)$$

where S^{ex} and S^{im} denote the attended representations guided by explicit and implicit priors, respectively. Note that although both encode action-relevant information, they may highlight different facets of the underlying motion. For

Methods	Guidance	Spatial		Object		Goal		Long		Avg.	
		SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$	SR $\uparrow$	Rank $\downarrow$
Diffusion Policy [11]	–	78.3	26	92.5	18	68.3	27	50.5	27	72.4	27
Octo [48]	–	78.9	25	85.7	26	84.6	20	51.1	26	75.1	25
CoT-VLA [62]	Visual	87.5	20	91.6	20	87.6	17	69.0	19	81.1	20
WorldVLA [8](256*256)	Visual	85.6	22	89.0	23	82.6	22	59.0	22	79.1	21
WorldVLA [8](512*512)	Visual	87.6	19	96.2	15	83.4	21	60.0	21	81.8	19
DreamVLA [60]	Visual	97.5	9	94.0	16	89.5	15	89.5	12	92.6	14
UniVLA [49]	Visual	95.4	14	98.8	4	93.6	12	94.0	6	95.5	10
F1 [36]	Visual	98.2	5	97.8	10	95.4	11	91.3	10	95.7	9
GE-Act [29]	Visual	98.2	5	97.6	12	95.8	9	94.4	5	96.5	7
TraceVLA [66]	Linguistics	84.6	24	85.2	27	75.1	26	54.1	24	74.8	26
OpenVLA [24]	Linguistics	84.7	23	88.4	24	79.2	23	53.7	25	76.5	24
UniAct [65]	Linguistics	77.0	27	87.0	25	77.0	25	70.0	18	77.8	23
SpatialVLA [41]	Linguistics	88.2	18	89.9	22	78.6	24	55.5	23	78.1	22
ThinkAct [17]	Linguistics	88.3	17	91.4	21	87.1	18	70.9	17	84.4	18
π_0 -FAST [39]	Linguistics	96.4	12	96.8	14	88.6	16	60.2	20	85.5	17
FPC-VLA [52]	Linguistics	87.0	21	92.0	19	86.2	19	82.2	15	86.9	16
SmolVLA [43]	Linguistics	93.0	16	94.0	16	91.0	14	77.0	16	88.8	15
GR00T-N1 [4]	Linguistics	94.4	15	97.6	12	93.0	13	90.6	11	93.9	13
π_0 [5]	Linguistics	96.8	11	98.8	4	95.8	9	85.2	14	94.1	12
GO-1 [6]	Linguistics	96.2	13	97.8	10	96.0	8	89.2	13	94.8	11
DD-VLA [28]	Linguistics	97.2	10	98.6	6	97.4	5	92.0	9	96.3	8
MemoryVLA [42]	Linguistics	98.4	4	98.4	7	96.4	7	93.4	7	96.7	6
$\pi_{0.5}$ [22]	Linguistics	98.8	2	98.2	9	98.0	3	92.4	8	96.9	5
OpenVLA-OFT [25]	Linguistics	97.6	8	98.4	7	97.9	4	94.5	4	97.1	4
VLA-Adapter [50]	Linguistics	97.8	7	99.2	2	97.2	6	95.0	3	97.3	3
Ours[◊]	Action	99.4	1	99.6	1	98.8	2	96.0	2	98.5	1
Ours	Action	98.6	3	99.0	3	99.4	1	97.0	1	98.5	1

Table 1. Comparison on the LIBERO benchmark. Our proposed approach is trained on the LIBERO dataset. $\diamond$ represents that the LLM backbone is frozen during training. All metrics are average success rates (%). The best results are highlighted in **bold**.

instance, explicit priors provide kinematic cues, whereas implicit priors capture latent action tendencies. Hence, to effectively combine these complementary guidance, we concatenate the two attended features and process them through self-attention fusion block, which integrates the priors into a unified representation $\bar{h}$:

$$\bar{h} = \text{Self-Attn}([S^{\text{ex}}; S^{\text{im}}]). \quad (11)$$

Eventually, the aggregated representation $\bar{h}$ is fed into $\pi_{\theta}^{\text{head}}$, which predicts the denoised action sequence $a_{t:t+H-1}$.

Training Objectives. The entire framework is optimized under a standard flow-matching mean-squared error (MSE) objective. The training losses consist of two parts, *i.e.*, flow-matching MSE for both Explicit Action Reasoner $\pi_{\theta}^{\text{ref}}$ and action head $\pi_{\theta}^{\text{head}}$, denoted as $\mathcal{L}_{\pi_{\theta}^{\text{ref}}}$ and $\mathcal{L}_{\pi_{\theta}^{\text{head}}}$, respectively. Hence, the overall objective is:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\pi_{\theta}^{\text{ref}}} + \lambda_2 \mathcal{L}_{\pi_{\theta}^{\text{head}}}, \quad (12)$$

where λ_1 and λ_2 are balance factors.

Teacher Forcing Stabilization. During training, the outputs of $\pi_{\theta}^{\text{ref}}$ can be unstable. To stabilize optimization, we compute Z^{ex} directly from ground-truth reference trajectories instead of from $\pi_{\theta}^{\text{ref}}$ predictions, preventing optimization interference to $\pi_{\theta}^{\text{head}}$. During inference, the model switches to a fully self-conditioned mode, where $\pi_{\theta}^{\text{ref}}$ autonomously generates the reference actions to guide $\pi_{\theta}^{\text{head}}$ in action prediction.

4. Experiments

In this section, we first outline the experimental setup in Sec. 4.1. Then, in Sec. 4.2, we evaluate our approach on three simulation benchmarks, followed by comprehensive ablation studies in Sec. 4.3. Moreover, we present real-world deployment results in Sec. 4.4 to evaluate real-world applicability.

4.1. Experimental Setup

Data Sources. For simulation experiments, we strictly follow the official training splits provided by the corresponding benchmark (LIBERO [32], LIBERO-Plus [15], and VLABench [59]), and train our models exclusively on their standard demonstration datasets without introducing any additional data. For the real-world setting, all demonstrations used for model training are collected on our own robotic platform. More details about data sources are introduced in Appendix A.

Implementation Details. We implement our approach upon $\pi_{0.5}$ [22]. Specifically, we adopt SigLIP [56] as the visual encoder, while the LLM backbone is instantiated as Gemma 2B architecture [3] with $N = 18$ layers and hidden size $d = 2048$. For frame processing, each input frame is resized to 224×224 prior to the visual encoder. Regarding the EAR, we employ a compact Transformer-based design composed of $N = 18$ layers. Concerning the IAR, each learnable query matrix Q_i is configured with a row dimen-

Methods	Guidance	Camera	Robot	Language	Light	Background	Noise	Layout	Avg.
<i>Zero-Shot Transfer</i>									
WorldVLA [8]	Visual	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
OpenVLA [24]	Linguistics	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
NORA [21]	Linguistics	2.2	37.0	65.1	45.7	58.6	12.8	62.1	39.0
UniVLA [7]	Linguistics	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
π_0 -Fast [39]	Linguistics	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
RIPT-VLA [44]	Linguistics	55.2	31.2	77.6	88.4	91.6	73.5	74.2	68.4
OpenVLA-OFT [25]	Linguistics	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
π_0^* [5]	Linguistics	61.0	40.8	63.5	89.3	84.1	80.1	76.4	69.4
$\pi_{0.5}^*$ [22]	Linguistics	75.8	79.4	83.3	95.5	95.0	89.6	87.0	85.7
Ours$^\diamond$	Action	68.9	80.3	84.1	95.6	93.1	81.5	88.3	83.6
Ours	Action	72.6	82.6	87.5	97.7	96.5	87.8	88.1	86.6
<i>Supervised Fine-Tuning</i>									
π_0^* [5]	Linguistics	79.6	21.1	72.5	84.7	86.2	68.3	69.4	67.4
$\pi_{0.5}^*$ [22]	Linguistics	70.3	41.7	81.1	97.3	94.6	71.8	84.9	75.7
Ours$^\diamond$	Action	91.2	62.5	80.3	95.1	91.5	88.3	84.9	84.1
Ours	Action	96.6	70.4	79.7	95.1	97.1	95.9	85.0	88.0

Table 2. Comparison on the LIBERO-Plus benchmark. Methods under *Zero-Shot Transfer* are trained on LIBERO dataset and directly evaluated on LIBERO-Plus. *Supervised Fine-Tuning* denotes models trained on the LIBERO-Plus training set. An asterisk (*) denotes results reproduced by utilizing officially released checkpoints, while $\diamond$ represents that the LLM backbone is frozen during training. The best results are highlighted in **bold**.

sion of $M = 1$. The reduced dimension in the downsampling strategy is set to $d' = 128$.

In terms of model training, unless explicitly specified, the horizon of predicted reference actions H^{ref} and action policy output H are fixed to 15 and 10, with action shift set to 2 and 1, respectively. To clarify, the action shift specifies the temporal interval relative to the expert demonstration. For instance, a shift of 1 yields frame-aligned predictions, whereas a shift of 2 skips one intermediate frame. We set the balance factors in training losses as $\lambda_1 = \lambda_2 = 0.5$.

Training Configuration. We adopt a unified set of training hyperparameters across all experiments unless explicitly specified. Concretely, the learning rate follows a cosine-decay schedule with a warm-up phase of 10K steps, a peak learning rate of $5e-5$, and a decay toward $5e-5$ over 10K steps. Optimization is performed with AdamW with gradient-norm clipping set to 1.0. An exponential moving average (EMA) of model parameters is maintained with a decay rate of 0.999. Regarding hardware settings, model training is performed on a single node equipped with 8 NVIDIA H100 GPUs using bfloat16 precision. And the inference is conducted on a single NVIDIA RTX 4090.

4.2. Simulation Experiments

In this section, we conduct the simulation evaluations across three benchmarks, *i.e.*, LIBERO [32], LIBERO-Plus [15], and VLABench [59], to comprehensively evaluate our approach’s performance and generalization capabilities under diverse task structures.

LIBERO Benchmark. We evaluate our approach on LIBERO benchmark, which targets four distinct robot ca-

pabilities: spatial awareness (Spatial), object manipulation (Object), goal completion (Goal), and long-horizon reasoning (Long). Each task suite consists of 10 tasks and provides 50 human-teleoperated demonstrations per task for policy training. The evaluation is conducted following the official evaluation protocol. For each task, the policy is evaluated over 50 trials, amounting to 2,000 total rollouts.

As reported in Table 1, the quantitative evaluation results demonstrate that our proposed approach outperforms existing methods across all tracks. Compared to previous state-of-the-art method $\pi_{0.5}$, our approach achieves a 1.6% absolute improvement in average. Notably, we observe a pronounced improvement on LIBERO-Long suite, where tasks require long-horizon manipulation with strict error control. Particularly, unlike Language- or Visual-CoT, whose intermediate reasoning remains abstract or indirect with respect to action execution, our proposed ACoT naturally operates in precise representation. Through leveraging actions as intermediate reasoning, the model feeds the action head with structured action guidance, which significantly enhances the robustness in long-horizon manipulation tasks.

LIBERO-Plus Benchmark. Built upon LIBERO, LIBERO-Plus is designed to systematically evaluate robotic policies under controlled distribution shifts. Concretely, LIBERO-Plus introduces 7 perturbation dimensions, *i.e.*, camera-viewpoints (Camera), robot-initial-states (Robot), language-variations (Language), lighting-conditions (Light), background-textures (Background), sensor-noise (Noise) and object-layout (Layout), exposing hidden failure modes under standard evaluations. Notably, LIBERO-Plus consists of 10,030 evaluation episodes, pro-

Methods	Guidance	In-dist.		Category		Commonsense		Instruction		Texture		Avg.	
		IS $\uparrow$	PS $\uparrow$	IS $\uparrow$	PS $\uparrow$	IS $\uparrow$	PS $\uparrow$	IS $\uparrow$	PS $\uparrow$	IS $\uparrow$	PS $\uparrow$	IS $\uparrow$	PS $\uparrow$
$\pi_0^\diamond$ [5]	Linguistics	67.8	62.7	44.0	33.6	54.9	43.0	58.0	38.7	50.6	42.5	55.0	44.1
$\pi_{0.5}^\diamond$ [22]	Linguistics	75.0	60.8	49.6	35.3	57.5	41.6	57.1	30.3	62.0	47.4	60.2	43.1
Ours$^\diamond$	Action	79.8	66.1	54.1	38.9	52.3	37.8	56.8	39.6	74.6	54.6	63.5	47.4

Table 3. Comparison on the VLABench benchmark. IS and PS represent Intention score and Progress score, respectively. All models are trained for 60K steps. $\diamond$ indicates that the LLM backbone is frozen during training. The best results are highlighted in **bold**.

Name	EAR	IAR	Spatial	Object	Goal	Long	Avg.
Baseline			98.8	98.2	98.0	92.4	96.9
#1	✓		99.0	99.4	98.0	96.6	98.3
#2		✓	99.2	99.2	98.2	95.6	98.1
#3	✓	✓	99.4	99.6	98.8	96.0	98.5

Table 4. Module ablations. The performance is gradually improved with the continuous addition of proposed methods.

viding statistically reliable evaluation.

We consider two evaluation protocols: (i) *Zero-Shot Transfer*, where policies trained on the LIBERO dataset are directly evaluated on LIBERO-Plus to assess generalization. (ii) *Supervised Fine-Tuning*, where models are directly optimized on LIBERO-Plus dataset. Technically, we follow evaluation configuration in LIBERO-Plus [15], *i.e.*, each episode is executed once without repeated rollouts.

As shown in Table 2, our method significantly boosts the policy’s performance, surpassing all previous methods in both settings. Specifically, under the *Zero-Shot* regime, our approach demonstrates pronounced robustness against distribution shifts such as robot initial-state perturbations (+3.2%) and language variations (+4.2%), where existing language-guided or vision-guided policies exhibit significant degradation. Furthermore, our method maintains exceptional performance under the *Supervised Fine-Tuning* setting, reaching an 88.0% average success rate. These results highlight the effectiveness of our action-space reasoning in improving generalization and robust policy learning.

VLABench Benchmark. Built on ManiSkill3 [45], VLABench is designed to benchmark both VLAs and VLMs on diverse robotic tasks. The standard evaluation is organized into 5 public tracks, *i.e.*, in-distribution, cross-category (category-level generalization), common-sense reasoning, semantic-instruction (language understanding), and unseen-texture (appearance robustness). Particularly, VLABench proposes Intention Score (IS) and Progress Score (PS) to evaluate robot policies.

In our context, we train π_0 , $\pi_{0.5}$, along with our method in a unified training setup on VLABench’s official training data. We present quantitative results in Table 3. Overall, our method achieves the best performance across both IS (63.5%) and PS (47.4%). Notably, under the challenging unseen-texture track, it delivers substantial gains, *i.e.*, +12.6% in IS and +7.2% in PS, indicating strong robustness to distribution shifts. Together, these results further confirm the effectiveness of our proposed approach.

Name	Action shift	Action horizon	Equi. horizon	Spatial	Object	Goal	Long	Avg.
Baseline	1	10	10	98.6	99.0	96.4	92.2	96.6
+EAR	1	10	10	99.4	99.4	98.8	95.0	98.2
	2	5	10	99.6	99.6	98.4	94.4	98.0
	1	30	30	99.2	99.2	97.6	95.6	97.9
	2	15	30	99.0	99.4	98.0	96.6	98.3
	2	30	60	99.4	99.0	98.2	95.0	97.9
	3	30	90	98.8	99.4	97.4	96.2	98.0

Table 5. Reference action parameter ablation. We observe that different reference-action configurations within EAR generally lead to performance improvements.

Methods	Spatial	Object	Goal	Long	Avg.
Baseline	98.8	98.2	98.0	92.4	96.9
Query	98.8	99.0	97.2	92.8	97.0
Attention Pooling	99.4	98.6	98.2	92.8	97.3
Downsample	99.2	99.2	98.2	95.6	98.1

Table 6. Comparison of KV-cache interaction strategies in IAR.

4.3. Ablation Study

We examine each component’s contribution via systematic ablation experiments on the LIBERO benchmark, which are shown in Table 4, Table 5, and Table 6. Note that we adopt $\pi_{0.5}$ as the “Baseline” method. More ablations in different benchmarks are in Appendix C.

EAR. As shown in Table 4, compared with the baseline, the experiment “#1” introduces the Explicit Action Reasoner (EAR) module into policy learning, which lifts the average success rate from 96.9% to 98.3%, demonstrating that the explicit action-space guidance benefits the robotic action sequence prediction. A plausible explanation is that EAR introduces an intermediate reference action sequence, which injects strong inductive bias on the behavior and thereby reduces ambiguity in mapping from observations to actions.

IAR. Analogously, with the Implicit Action Reasoner (IAR) module added in “#2”, the average success rate increases from 96.9% to 98.1%. This gain suggests that exploiting the implicit action distribution encoded in vision–language representations can also provide effective guidance for policy learning. This performance gain can be partly attributed to the fact that IAR distills action-related clues implicitly encoded within the vision–language backbone, which potentially reflects the distribution of feasible actions. Such priors encourages the policy to remain closer to coherent, task-consistent behavioral patterns.

EAR + IAR. In Table 4, experiment “#3” incorporates both

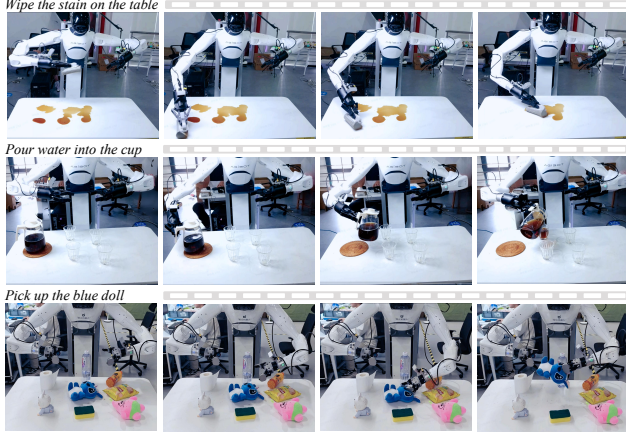


Figure 3. Visualization of three manipulation tasks in real world.

EAR and IAR, achieving the highest average success rate of 98.5%. The consistent improvements demonstrate that explicit action guidance and implicit action cues extracted from VLM’s key-value cache are complementary, jointly providing stronger guidance for accurate action prediction.

Reference Action Configurations. To further examine the effect of explicit action references in EAR, we investigate different settings of action shift and action horizon, as summarized in Table 5. We observe various parameter combinations consistently bring improvements over the baseline, indicating that providing action cues is broadly beneficial for policy learning. Besides, we find that shorter horizons combined with moderate shifts tend to produce relatively stronger gains. These observations offer further insight into how explicit action guidance influence policy learning.

KV-cache Interaction Strategies. We compare three strategies for extracting action-relevant cues from VLM’s key-value cache within IAR module, as presented in Table 6. Concretely, “Query” method uses learnable queries to attend to VLM’s original key-value cache. “Attention Pooling” method forms a pooled query by averaging key-value cache and then applies cross-attention operation. “Downsample” method first downsamples VLM’s key-value cache and then aggregates them using learnable matrix.

As shown in Table 6, all three variants outperform the baseline, indicating that extracting implicit action cues from VLM benefits policy learning. Notably, the “Downsample” strategy achieves the best performance, suggesting that VLM’s features may contain noisy information for action prediction. This also highlights the importance of designing appropriate interaction mechanisms to align vision-language and action.

4.4. Real-World Deployment

To further validate the effectiveness of our framework, we conduct extensive real-world experiments on the AgiBot G1 robot. We consider three manipulation tasks, *i.e.*, “Wipe

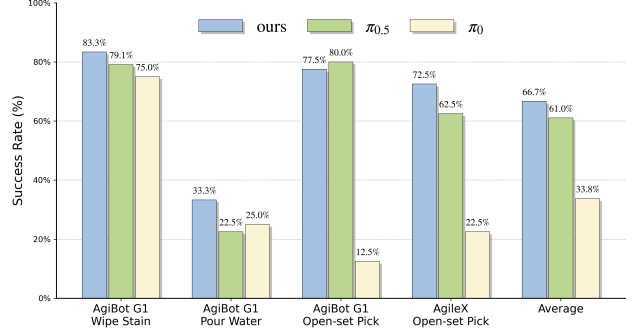


Figure 4. Evaluation results of real-world experiments.

Stain”, “Pour Water”, and “Open-set Pick”, which respectively assess contact-rich manipulation, fine-grained object handling, and instruction-following abilities.

Specifically, as visualized in Fig. 3, (i) the “Wipe Stain” task requires the robot to pick up a sponge from the table and wipe away the stain until the surface is clean. (ii) The “Pour Water” task requires the robot to grasp the kettle by its handle, locate the target cup, pour water into it without causing overflow, and finally return the kettle to the table in a stable manner. (iii) The “Open-set Pick” task instructs the robot to pick up the correct tabletop object according to given natural-language command. Additionally, to examine the cross-embodiment adaptability, we also perform the “Open-set Pick” task on the AgileX robotic platform. More details about training and evaluation are provided in the supplementary material.

As shown in Fig. 4, our approach achieves consistently higher average success rates than both $\pi_{0.5}$ and π_0 , *i.e.*, 66.7% against 61.0% and 33.8%. These results demonstrate that the proposed framework maintains effectiveness under real-world sensing conditions. Moreover, the aligned improvements observed on both AgiBot G1 and AgileX also indicate that our method exhibits adaptability across different robotic embodiments.

5. Conclusion

In this work, we address the fundamental semantic-kinematic gap in modern robotic policies by proposing a new paradigm: Action Chain-of-Thought (ACoT). We argue that for physically grounded intelligence, deliberation should occur not in the abstract space of language or vision, but directly in the kinematically grounded space of actions. We materialize this concept in our ACoT-VLA framework, which leverages two synergistic modules—an Explicit Action Reasoner (EAR) and an Implicit Action Reasoner (IAR)—to generate and fuse both explicit trajectory plans and implicit behavioral priors. This action-centric guidance mechanism creates a direct, information-rich conduit between high-level intent and low-level motor control. Extensive experiments across multiple simulation and real-

world benchmarks demonstrate that our proposed approach yields state-of-the-art performance. Through shifting the locus of reasoning from perception to action, our work not only provides a more effective and grounded method for robotic policy learning, but also opens a new avenue for research into more structured, interpretable, and capable embodied agents. We believe that learning to “think” in the language of actions is a critical step towards developing the next generation of generalist robots.

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A. Dataset Description

In this section, we present a comprehensive characterization of the benchmark datasets and the custom-collected data used for model training in our experiments. We systematically report key statistics, including the total number of episodes, frame counts, and other relevant properties, which is summarized in Table 7 below:

Type	Dataset	Embodiment	DoF	Episodes	Frames	FPS
Simulation	LIBERO	Franka	7	1,693	273,465	10
	LIBERO-Plus	Franka	7	14,347	2,238,036	20
	VLABench	Franka	7	4,713	528,398	10
Real-World	Wipe Stain	AgiBot G1	22	177	356,316	30
	Pour Water	AgiBot G1	22	1,821	5,062,506	30
	Open-set Pick	AgiBot G1	22	1,936	219,824	30
	Open-set Pick	AgileX	14	962	251,283	30

Table 7. Dataset statistics.

Simulation Benchmarks. We utilize three publicly released simulation datasets, *i.e.*, LIBERO [32], LIBERO-Plus [15], and VLABench [59]. Specifically, the LIBERO dataset contains 1,693 episodes and 273,465 frames, recorded at a fixed 10 Hz. Its demonstrations exhibit relatively uniform trajectory lengths and smooth motion patterns, making it widely adopted benchmark in community.

However, due to the increasing performance saturation observed on LIBERO, LIBERO-Plus is recently introduced to provide a more challenging and diversified evaluation setting. LIBERO-Plus provides 14,347 episodes and 2,238,036 frames, captured at 20 Hz. In contrast to the homogeneous trajectories in LIBERO, LIBERO-Plus explicitly emphasizes a perturbation-oriented design. The demonstrations display substantially larger variations in motion magnitude and camera-robot viewpoint configuration. These characteristics make it a more suitable benchmark for evaluating policy generalization under structured distribution shifts.

Besides these two datasets, we further benchmark our method on VLABench, whose training set includes 4,713 episodes and 528,398 frames, recorded at 10 Hz, which requires a higher level of visual and physical understanding from the policy.

Real-World Experiment. For real-world deployment, we collect demonstrations across 3 tasks, *i.e.*, Wipe Stain, Pour Water, and Open-set Pick, as shown in Table 7.

The “Wipe Stain” dataset contains 177 episodes with 356,316 frames, characterized by dense tool-surface contact and fine-grained force control. The “Pour Water” dataset includes 1,821 episodes and 5,062,506 frames. Its large scale stems from the task’s long-horizon and multi-stage nature. Regarding the “Open-set Pick” task, the AgiBot G1 subset provides 1,936 episodes with 219,824 frames, while the AgileX subset offers 962 episodes with 251,283 frames, both featuring diverse tabletop layouts and natural-language instructions.

Task	Action Space	Action Horizon	State	Batch Size	Training Step
LIBERO	Delta EEF	10	×	128	40K
LIBERO-Plus	Delta EEF	10	×	128	100K
VLABench	Abs EEF	10	✓	128	60K
Wipe Stain	Abs Joint	30	✓	128	50K
Pour Water	Abs Joint	30	✓	128	240K
Open-set Pick	Abs Joint	30	✓	128	50K
Open-set Pick [†]	Abs Joint	30	✓	128	50K

Table 8. Training details. Note that the “Open-set Pick[†]” task is performed on AgileX platform.

B. Training & Evaluation Details

Training Details. We describe the task-specific training configurations, *e.g.*, action space and state usage, for better understanding.

As presented in Table 8, for the LIBERO and LIBERO-Plus suites, the policy is trained using delta end-effector control (Delta EEF) with an action horizon of 10 steps. In particular, no privileged state information is provided during training. We utilize a global batch size of 128 and train the policies for 40K and 100K steps, respectively. Similarly, we train our models in VLABench for 60K steps, while adopting state input and absolute end-effector (Abs EEF) actions to align the benchmark’s control convention.

In terms of the real-world tasks, we utilize Abs Joint control with a longer action horizon of 30. Unlike the simulator benchmarks, these tasks additionally provide structured robot state observations to improve robustness under real-world sensing and actuation noise. Our models are trained for 50K, 240K, and 50K steps, in “Wipe Stain”, “Pour Water”, and “Open-set Pick” tasks, respectively, with same batch size of 128.

Evaluation Details. Next, we illustrate the evaluation protocols and success criteria for all real-world tasks. Each task is assessed using fixed and repeatable initializations to ensure reproducibility and reduce environmental variance.

Concretely, in terms of the “Wipe Stain” task, we predefine three initial sponge poses. For each pose, the robot is required to clean stains placed at four distinct table locations. Every configuration is executed twice, resulting in 24 trials in total. A trial is considered successful if the robot grasps the sponge and removes the stain from the specified location.

As for the “Pour Water”, we standardize six predefined relative configurations between the bottle and the glass. Then, each configuration is executed two times. A trial is counted as successful if the robot lifts the bottle, pours water into the cup, and places the bottle back onto the coaster. Note that minor spillage of water when pouring is allowed.

Eventually, regarding the “Open-set Pick” task, we initialize ten object arrangements on the table, containing both in-distribution and out-of-distribution instances. In each arrangement, the robot is instructed to grasp a specified target object using either its left or right arm, as indicated by the

Name	EAR	IAR	Camera	Robot	Language	Light	Background	Noise	Layout	Avg.
Baseline			70.3	41.7	81.1	97.3	94.6	71.8	84.9	75.7
#1	✓		88.7	63.5	80.4	94.0	90.2	89.5	84.2	83.7
#2		✓	80.7	48.7	82.6	97.7	90.9	84.3	86.0	80.4
#3	✓	✓	91.2	62.5	80.3	95.1	91.5	88.3	84.9	84.1

Table 9. Module ablations on LIBERO-Plus benchmark. The performance is gradually improved with the addition of proposed methods. Note that models are directly optimized on LIBERO-Plus dataset, with the LLM backbone frozen during training.

Name	Action Head		EAR		LIBERO					LIBERO-Plus							
	Param.	Denoise	Param.	Denoise	Spatial	Object	Goal	Long	Avg.	Camera	Robot	Language	Light	Background	Noise	Layout	Avg.
Baseline	300M	10	-	-	98.6	99.0	96.4	92.2	96.6	70.3	41.7	81.1	<u>97.3</u>	94.6	71.8	<u>84.9</u>	75.7
#1	600M	10	-	-	97.6	98.4	97.8	<u>96.4</u>	97.6	68.7	44.8	<u>83.1</u>	96.4	92.7	66.6	84.1	74.9
#2	600M	20	-	-	97.8	98.8	<u>98.0</u>	95.2	97.5	70.0	44.8	82.7	97.6	93.1	66.7	83.2	75.1
#3	300M	5	300M	5	98.6	99.6	97.8	95.4	<u>97.9</u>	<u>88.2</u>	<u>62.4</u>	81.5	95.0	91.5	88.6	85.3	83.9
#4	300M	10	300M	10	<u>99.0</u>	<u>99.4</u>	<u>98.0</u>	96.6	98.3	88.7	63.5	80.4	94.0	90.2	89.5	84.2	<u>83.7</u>
#4	300M	10	300M	10	<u>99.0</u>	<u>99.4</u>	<u>98.0</u>	96.6	98.3	88.7	63.5	80.4	94.0	90.2	89.5	84.2	<u>83.7</u>
#5	300M	10	150M	10	99.2	99.2	97.8	94.2	97.6	86.4	54.3	81.7	92.2	91.4	<u>89.1</u>	82.1	81.7
#6	300M	10	250M	10	<u>99.0</u>	98.2	98.6	94.2	97.5	87.2	59.7	81.1	95.0	<u>93.7</u>	87.4	83.5	83.1
#7	300M	10	500M	10	98.4	<u>99.4</u>	96.6	94.2	97.0	80.8	57.6	84.1	95.6	92.1	79.8	83.7	80.9

Table 10. Effects of parameters and denoise steps on policy performance. Note that the IAR module is not added in this experiment. The evaluation protocol in LIBERO-Plus is *Supervised Fine-Tuning*, i.e., models are directly optimized on LIBERO-Plus dataset. The LLM backbone is frozen during training. The best results are highlighted in **bold**, and the second-best results are underlined.

instruction. Each arm–object pair is evaluated twice, resulting in 40 trials overall. A trial is deemed successful if the robot grasps the instructed object with the correct arm.

Across all tasks, evaluations are carried out by trained operators with substantial prior testing experience, and success rates are computed as the proportion of successful trials relative to the total number of executed attempts.

C. More Experimental Results

In this section, we provide additional quantitative experiments to substantiate the effectiveness of our proposed approach and to empirically uncover several insightful phenomena. Specifically, the experimental analyses comprise four parts: (1) ablation study conducted on the LIBERO-Plus benchmark in Table 9, (2) an investigation of how the parameter sizes of the Action Head and Explicit Action Reasoner (EAR), as well as the number of denoising steps, influence policy performance in Table 10, (3) evaluation of proposed approach’s sim2real capability based on Genie-Sim 3.0 [53] in Table 11, and (4) a comparative study examining the relationship among inference latency, model size, and performance in Table 12. Note that we adopt $\pi_{0.5}$ [22] as the baseline method, denoted as “Baseline”.

Module Ablation. As shown in Table 9, incorporating the proposed reasoning modules consistently improves policy performance on the LIBERO-Plus benchmark. Adding

the EAR module, i.e., experiment “#1”, yields a clear gain over the baseline, increasing the average success rate from 75.7% to 83.7%. This improvement can be attributed to EAR’s ability to generate an explicit reference action trajectory, which significantly reduces the ambiguity in mapping complex visual or linguistic observations to low-level actions, such as camera shifts and background changes. Meanwhile, incorporating only the IAR (“#2”) also improves the performance from 75.7% to 80.4%, indicating that decoding the latent action-related semantics within the vision–language backbone provides useful behavioral priors. Finally, combining EAR and IAR (“#3”) achieves the highest success rate of 84.1%, demonstrating their complementary effects, i.e., EAR provides explicit motion guidance, while IAR supplies dense representation-level priors.

Effect of Model Scaling & Denoising Budget. Then, we analyze the superiority of our method by comparing settings with matched total model parameters and denoising steps. As shown in Table 10, firstly, we enlarge the model size of the action head and increase the number of denoising steps in experiments “#1” and “#2”, to construct fair baselines for subsequent comparison. We observe a preliminary observation, i.e., increasing the model size or denoising steps does not reliably enhance performance. Specifically, compared with the baseline, while “#1” improves performance on the LIBERO benchmark, it simultaneously drops on LIBERO-Plus. Next, comparing “#1” and “#2” reveals that further in-

Tasks	Simulation		Real-World	
	$\pi_{0.5}$	Ours	$\pi_{0.5}$	Ours
Select Color	86.0	98.8	85.0	94.0
Recognize Size	93.0	96.0	94.0	94.0
Grasp Targets	71.7	68.0	70.8	75.0
Organize Objects	52.0	74.0	60.0	68.4
Avg.	75.7	84.2	77.5	82.9

Table 11. Experimental results on Genie Sim 3.0 Simulation and Real-world Transfer. The best results are highlighted in **bold**.

creasing denoising steps yields only negligible fluctuations.

Subsequently, we incorporate the EAR module under fully matched overall parameterization and denoising budgets. Concretely, in both comparison pairs, “#1” with “#3” and “#2” with “#4”, we consistently observe notable performance improvements on both benchmarks, once the EAR module is introduced. This indicates that the performance gains originate from our proposed action chain-of-thought. The proposed mechanism supplies explicit reference actions that effectively mitigate the intrinsic instability of action prediction, especially under challenging external perturbations, as shown in the LIBERO-Plus, enabling a more reliable and grounded generalist robotic policy.

Effect of EAR Scale. Moreover, we investigate how various scale of the EAR module influences action prediction fidelity. To isolate the effect of EAR, we keep the action head parameters and the denoising schedule strictly fixed, while scaling the EAR module to 150M, 250M, 300M, and 500M parameters via adjusting hidden size. As presented in Table 10, through the comparison across experiments “#4”, “#5”, “#6”, and “#7”, we find that although all EAR-equipped variants outperform non-EAR baselines on both benchmarks, the performance trend is non-monotonic. Applying moderate EAR scales, *e.g.*, 300M, yields the greatest improvement. Particularly, as evidenced in “#7” in Table 10, when the parameter of EAR module even exceeds that of the action head, we observe a marked drop in performance. We attribute this degradation to the tendency of an over-parameterized EAR to overfit spurious correlations during training. Therefore, it generates reference action trajectories that are systematically biased, which ultimately misdirect the action head toward suboptimal predictions.

Sim To Real Evaluation. To further assess the sim2real transferability of proposed method, we conduct evaluations on the Genie-Sim 3.0 benchmark [53]. It comprises 4 challenging tasks: picking objects based on specified colors, sizes, or categories, as well as a complex tabletop organization task. Technically, the model is trained on the officially released simulation-based datasets. Then, it is deployed and evaluated in both simulation and real-world environments.

As illustrated in Table 11, our approach consistently outperforms the baseline across both domains. Specifically, it achieves 84.2% in simulation and 82.9% in real-world

Name	EAR	IAR	Param.	Latency	LIBERO Avg. SR	LIBERO-Plus Avg. SR
Baseline			3.35B	91ms	96.9	75.7
#1	✓		3.80B	110ms	98.3	83.7
#2		✓	3.36B	93ms	98.1	80.4
#3	✓	✓	3.81B	112ms	98.5	84.1

Table 12. Ablation experiment on model efficiency and performance. Note that the evaluation protocol in LIBERO-Plus is *Supervised Fine-Tuning*.

settings, representing absolute improvements of 8.5% and 5.4%, respectively. Notably, our method exhibits minimal performance degradation during the sim-to-real transition, which we attribute to the fundamental nature of ACoT paradigm. Specifically, while visual domain gap persists between synthetic and real-world observations, the underlying kinematically grounded action guidance remains consistent. By shifting the locus of reasoning from the perceptual space to the action space, our model extracts task-relevant motion priors that are invariant to low-level visual perturbations, effectively enhancing policy’s sim2real capability.

Latency Analysis. In Table 12, we further examine the inference efficiency of our approach in terms of both parameter count and end-to-end latency. As additional reasoning modules are introduced, we observe a slight increase in latency. Incorporating the EAR module raises latency from 91ms to 110ms, while adding the IAR module introduces only an additional 2ms. However, this marginal overhead is outweighed by the substantial improvement, which reflects a favorable trade-off.

D. Limitations & Future Works

In this section, we discuss the limitations existing in our work and promising directions for future research.

Although our proposed action chain-of-thought (ACoT) substantially boosts policy performance, our framework still exhibits several constraints. The reasoning modules introduce additional computational cost, which, while relatively modest compared to the performance gains, may pose challenges for deployment on resource-constrained robotic platforms. Besides, another limitation stems from the fact that the prevailing action representation in the community is implemented as action chunks, *i.e.*, sequences of low-level control commands such as joint angles or end-effector poses. While such representations faithfully encode the executed motions, they lack explicit geometric structure that would facilitate higher-level spatial reasoning, such as object-centric coordination and contact geometry. Hence, the potential of ACoT reasoning may not be fully unleashed. Enriching action representations with spatially grounded information to enable ACoT to operate in geometrically interpretable 3D space, constitutes an interesting and promising avenue for future exploration.

E. LLM Usage Statement

In this paper, we employ Large Language Models (LLMs) solely for minor linguistic refinement during the manuscript preparation stage, such as correcting grammatical errors. None of the technical content, implementation details, or experimental results were generated by LLMs.